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[22ECCT502]

FIFTH SEMESTER B.Tech EXAMINATION, DECEMBER – 2024

ANTENNAS FOR MOBILE SYSTEMS

Time: 3 Hours **Maximum Marks: 100**

Instructions: i. Missing data may be suitably assumed.

ANSWER ALL QUESTIONS

		PART – A	**5 x 2 = 10**		
1.	**a.**	What factors affect signal propagation in mobile communication systems?	L2	CO1	**02**
	b.	What is the difference between uplink (UL) and downlink (DL) in mobile communication systems?	L2	CO2	**02**
	c.	How do empirical path loss models differ from physical models?	L2	CO3	**02**
	d.	What are megacells, and how do they differ from traditional macrocells?	L2	CO4	**02**
	e.	Explain base station antennas are typically mounted high above the ground?	L2	CO5	**02**
		PART – B	**6 x 5 = 30**		
2.	**a.**	How does signal propagation in indoor environments pose challenges for wireless communication, and how can these challenges be addressed?	L2	CO1	**05**
		OR			
	b.	Illustrate few applications that demand innovative antenna design.	L2	CO1	**05**
3.	**a.**	Explain the concept of the two-wave model on flat ground and provide necessary equations.	L2	CO2	**05**
		OR			
	b.	Explain and classify the multipath propagation models for Narrowband, Wideband systems with necessary equations.	L2	CO2	**05**
4.	**a.**	Explain the key parameters that define macrocells in mobile communication systems.	L3	CO3	**05**
		OR			
	b.	Explain rooftop diffraction for a macro cell system with necessary equations and diagrams.	L3	CO3	**05**
5.	**a.**	Define micro-cells, pico-cells, and femto-cells and what roles does it play in enhancing modern mobile systems.	L2	CO4	**05**
		OR			

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	b.	Explain the concept of shadowing and fast fading with respect to megacell system.	L2	CO4	**05**
6.	**a.**	Discuss the effects of multipath in a pico-cell system.	L2	CO4	**05**
		OR			
	b.	Explain empirical narrowband models with respect to megacell system.	L2	CO4	**05**
7.	**a.**	Explain the significance of compact antenna design in mobile terminals.	L2	CO6	**05**
		OR			
	b.	What is radio zone design in the context of base station antennas, and why is it important for optimizing coverage and capacity?	L2	CO6	**05**
		PART – C	**6 x 10 = 60**		
8.	**a.**	Explain the role of antennas in wireless communication and their impact on network performance.	L2	CO1	**10**
		OR			
	b.	Discuss the current trends in mobile communication and their impact on mobile devices.	L2	CO1	**10**
9.	**a.**	Explain the different frequency bands used in mobile communication systems and their applications.	L2	CO2	**10**
		OR			
	b.	Explain Spatio-Temporal Propagation Channel Model and its significance in understanding signal behavior in mobile communication systems.	L2	CO2	**10**
10.	**a.**	Illustrate the following physical path loss models for a macro cell system with necessary equations and diagrams. i. Allsebrook and Parsons Model ii. Ikegami Model	L2	CO3	**10**
		OR			
	b.	Illustrate the following empirical path loss models for a macro cell system with necessary equations and diagrams. i. Okumura Hata Model ii. Ibrahim and Parsons Model	L2	CO3	**10**

11.	**a.**	Explain the following with respect to mega-cell system. i. Shadowing and Fast Fading ii. Local Shadowing Effects iii. Local Multipath Effects	L2	CO4	**10**
		OR			
	b.	Illustrate the following empirical path loss models for a pico-cell system with necessary equations and diagrams. i. Wall and Floor Factor Model ii. COST 231 Multiwall Model	L2	CO4	**10**
12.	**a.**	Describe the different types of multiband antennas utilized in mobile devices and highlight their benefits?		CO5	**10**
		OR			
	b.	Define diversity and explain its impact on base station antennas.	L2	CO5	**10**
13.	**a.**	Discuss the basic techniques used in mobile terminal antenna design and their significance.	L2	CO6	**10**
		OR			
	b.	Describe the key parameters involved in designing a mobile terminal antenna.	L2	CO6	**10**